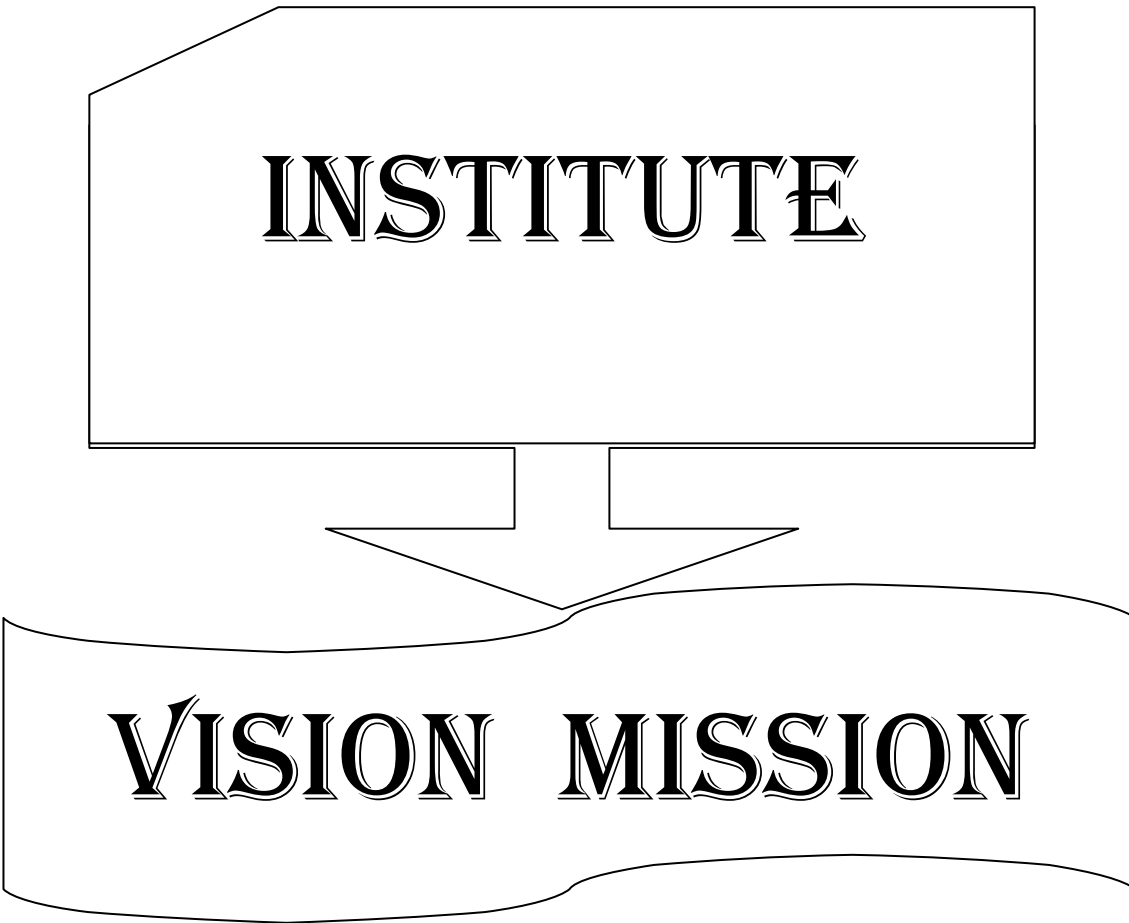


Academic Year
2021-22
VII SEM CSE
Handbook
(V18 Regulation)

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INSTITUTE

VISION MISSION

INSTITUTE VISION AND MISSION

VISION

To be a premier technological institute striving for excellence with global perspective and commitment to the nation.

MISSION

- To produce engineering graduates of professional quality and global perspective through Learner Centric Education.
- To establish linkages with government, industry and research laboratories to promote R&D activities and to disseminate innovations.
- To create an eco-system in the institute that leads to holistic development and ability for life-long learning.

DEPARTMENT

VISION

MISSION

DEPARTMENT

VISION AND MISSION

Vision:

- To evolve as a centre of academic and research excellence in the area of Computer Science and Engineering.

Mission :

- To utilize innovative learning methods for academic improvement.
- To encourage higher studies and research to meet the futuristic requirements of Computer Science and Engineering.
- To inculcate Ethics and Human values for developing students with good character

**PROGRAM
EDUCATIONAL
OBJECTIVES,
PROGRAM OUTCOMES
& PROGRAM
SPECIFIC
OUTCOMES**

Program Educational Objectives (PEOs)

Graduates of this programme will :

PEO 1: Adapt to evolving technology.

PEO 2: Provide optimal solutions to real time problems.

PEO 3: Demonstrate his/her abilities to support service activities with due consideration for Professional and Ethical Values.

Programme Specific Outcomes (PSO s):

A graduate of the Computer Science and Engineering Program will be able to:

PSO 1: Use Mathematical Abstractions and Algorithmic Design along with Open Source Programming tools to solve complexities involved in Programming. [K3]

PSO 2: Use Professional engineering practices and strategies for development and maintenance of software. [K3]

Program Outcomes (POs):

Computer Science Engineering Graduates will be able to:

1. **Engineering knowledge:** Apply the knowledge of Mathematics, Science, Engineering Fundamentals and Concepts of Computer Science Engineering to the solution of complex Engineering problems. [K3]
2. **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of Mathematics, Natural Sciences and Computer Science. [K4]
3. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specific needs with appropriate consideration for the public health and safety, and the cultural, societal and environmental considerations. [K5]
4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions. [K5]
5. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex Engineering activities with an understanding of the limitations. [K3]
6. **The Engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional Engineering practice. [K3]
7. **Environment and sustainability:**
Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for, sustainable development. [K3]
8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the Engineering practice. [K3]
9. **Individual and teamwork:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings. [K6]
10. **Communication:** Communicate effectively on complex Engineering activities with the Engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions. [K2]
11. **Project management and finance:** Demonstrate knowledge and understanding of the Engineering and Management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments. [K6]
12. **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change. [K1]

ACADEMIC CALENDAR

✉ : principal@srivasaviengg.ac.in
svec.as@gmail.com



☎ : 08818- 284344, 355

SRI VASAVI ENGINEERING COLLEGE (AUTONOMOUS)

(Sponsored by Sri Vasavi Educational Society)

(Approved by AICTE, New Delhi & Permanently affiliated to JNTUK, Kakinada)

(Accredited by NAAC with 'A' Grade, Recognized by UGC under section 2(f) & 12(B))

(NBA Accreditation to B.Tech., EEE, CSE, ME and ECE Branches for 3 Years)

Pedatadepalli, TADEPALLIGUDEM – 534 101. W.G.Dist. (A.P)

Principal's Office
Date: 04-09-2021

Academic Calendar **For B.Tech VII sem & VIII sem, Academic Year 2021-22** **(2018 Admitted Batch)**

VII Semester			
Description	From	To	Weeks
Commencement of Class Work	01.09.2021		
I Unit of Instructions	01.09.2021	16.10.2021	7 W
I Mid Examinations	18.10.2021	23.10.2021	1 W
II Unit of Instructions	25.10.2021	11.12.2021	7 W
II Mid Examinations	13.12.2021	18.12.2021	1 W
Preparation & Practicals	20.12.2021	25.12.2021	1 W
End Examinations	27.12.2021	08.01.2022	2 W
Commencement of VIII Semester Class Work	17.01.2022		
VIII Semester			
I Unit of Instructions	17.01.2022	05.03.2022	7 W
I Mid Examinations	07.03.2022	12.03.2022	1 W
II Unit of Instructions	14.03.2022	30.04.2022	7 W
II Mid Examinations	02.05.2022	07.05.2022	1 W
Preparation & Practicals	09.05.2022	14.05.2022	1 W
End Examinations	16.05.2022	28.05.2022	2 W
Commencement of Next Year Class Work			
Note : Calendar is prepared with 8 hrs/day hence 7 weeks per instruction period. Internal Examinations shall be conducted during the instructional days.			


PRINCIPAL

Copy to : ALL

Vision

To be a premier technological institute striving for excellence with global perspective and commitment to the nation.

Mission

- To produce Engineering graduates of professional quality and global perspective through learner-centric education.
- To establish linkages with government, industry and Research laboratories to promote R&D activities and to disseminate innovations.
- To create an eco-system in the institute that leads to holistic development and ability for life-long learning.



SRI VASAVI ENGINEERING COLLEGE (Autonomous)

Pedatadepalli, TADEPALLIGUDEM-534 101, W.G. Dist.

Department Of Computer Science & Engineering (Accredited by NBA)



Class: VII SEM

Section – A

Class Coordinator: Mrs. G Prashanthi

Room: PG-302

Periods	1	2	3	4	1:00P M 2:00P M	5	6	7
Time Day	(09.30 AM- 10.30 AM)	(10.30 AM- 11.20 AM)	(11.20 AM- 12.10 PM)	(12.10 PM- 01.00 PM)		(02.00 PM- 02.50 PM)	(02.50 PM- 03.40 PM)	(03.40 PM- 04.30 PM)
Mon	HCI	HCI	MS	AJWT	LUNCH BREAK	DS	DS	EPC
Tue	AJWT	DS	EPC	HCI		MS	AJWT	MS
Wed	DS	MS	AJWT	MS		EPC	HCI	LIB/SPORTS
Thu	EPC	HCI	DS	AJWT		AJWT LAB		
Fri	Project Work (Part-A)					Project Work (Part-A)		

Class: VII SEM

Section – B

Class Coordinator: Mrs.D S L Manikanteswari

Room: PG-303

Period s	1	2	3	4	1:00P M 2:00P M	5	6	7
Time Day	(09.30 AM- 10.30 AM)	(10.30 AM- 11.20 AM)	(11.20 AM- 12.10 PM)	(12.10 PM- 01.00 PM)		(02.00 PM- 02.50 PM)	(02.50 PM- 03.40 PM)	(03.40 PM- 04.30 PM)
Mon	EPC	AJWT	DS	MS	LUNCH BREAK	MS	AJWT	HCI
Tue	HCI	HCI	MS	AJWT		EPC	DS	DS
Wed	AJWT	EPC	DS	HCI		AJWT LAB		
Thu	DS	AJWT	MS	MS		HCI	EPC	LIB/SPORTS
Fri	Project Work (Part-A)					Project Work (Part-A)		

Class: VII SEM

Section – C

Class Coordinator: Mr. G.Sriram Ganesh Room: PG-401

Periods	1	2	3	4	1:00P M 2:00P M	5	6	7
Time Day	(09.30 AM- 10.30 AM)	(10.30 AM- 11.20 AM)	(11.20 AM- 12.10 PM)	(12.10 PM- 01.00 PM)		(02.00 PM- 02.50 PM)	(02.50 PM- 03.40 PM)	(03.40 PM- 04.30 PM)
Mon	EPC	MS	AJWT	DS	LUNCH BREAK	MS	EPC	HCI
Tue	AJWT	HCI	DS	LIB/SPORTS		AJWT LAB		
Wed	DS	AJWT	MS	EPC		HCI	DS	AJWT
Thu	HCI	MS	MS	DS		AJWT	EPC	HCI
Fri	Project Work (Part-A)					Project Work (Part-A)		

Class: VII SEM

Section – D

Class Coordinator: Mrs.M.S.Radha Mangamani

Room: PG-402

Periods	1	2	3	4	1:00P M 2:00P M	5	6	7
Time Day	(09.30 AM- 10.30 AM)	(10.30 AM- 11.20 AM)	(11.20 AM- 12.10 PM)	(12.10 PM- 01.00 PM)		(02.00 PM- 02.50 PM)	(02.50 PM- 03.40 PM)	(03.40 PM- 04.30 PM)
Mon	AJWT	DS	CES	HCI	LUNCH BREAK	AJWT LAB		
Tue	DS	MS	CES	AJWT		MS	HCI	CES
Wed	HCI	HCI	DS	MS		AJWT	CES	MS
Thu	CES	AJWT	AJWT	HCI		MS	DS	DS
Fri	Project Work (Part-A)					Project Work (Part-A)		

Staff Details:

S. No.	Course Code	Course Name	Sections			
			A	B	C	D
1.	V18CST27	Advanced Java and Web Technologies	Dr. K Shirin Bhanu	Dr. K Shirin Bhanu	Mrs.M.S.Radha Mangamani	Mrs.M.S.Radha Mangamani
2.	V18MBT52	Management Science (MS)	B.Aruna	P.Bharath Kumar	Dr.K.Pulla Rao	K.Murali Krishna
3.	V18CST31	Elective – III Human Computer Interaction (HCI)	Mrs. G Prashanthi	Mrs. G Prashanthi	Mr.A.Rajesh	Mrs. G Prashanthi
4.	V18CST32	Elective – IV Distributed Systems (DS)	Mrs.D S L Manikanteswari	Mrs.D S L Manikanteswari	Mr.G.Sriram Ganesh	Mr.G.Sriram Ganesh
5.	V18CEEEOE03	Open Elective – II Environmental Pollution and Control (EPC)	Mr. S Nagarjuna	Mr. S Nagarjuna	Mr. T Ramakrishna	Mr. A L S K S V D Gupta
6.	V18CSL10	Advanced Java and Web Technologies Lab (AJWT Lab)	Dr. K Shirin Bhanu / Mrs.D S L Manikanteswari	Dr. K Shirin Bhanu / Mrs.D S L Manikanteswari	Mrs.M.S.Radha Mangamani / Mr.G.Sriram Ganesh	Mrs.M.S.Radha Mangamani / Mr.G.Sriram Ganesh
7.	V18CSP01	Project Work (Part-A)	Mr.M Satyanarayana Reddy	Mrs.N Hiranmayee	Mr. M S Kumar Reddy	Dr.PLaxmikanth



Head of the Department

COURSE STRUCTURE

VII Semester

S.No.	VII - Semester						
	Course Code		Course	L	T	P	C
1	V18CST27	PCC	Advanced Java and Web Technologies	3	0	0	3
2	V18MBET52	HSS	Management Science	3	0	0	3
3	Elective – III						
	V18CST28	PEC	1. Advanced Operating Systems	3	0	0	3
	V18CST29		2. Statistics with R Programming				
	V18CST30		3. Information Retrieval Systems				
	V18CST31		4 Human Computer Interaction				
4	Elective – IV						
	V18CST32	PEC	1.Distributed Systems	3	0	0	3
	V18CST33		2.Scripting Languages				
	V18CST34		3.Deep Learning				
	V18CST35		4.Social Networks and semantic web				
5	Open Elective – II (Interdisciplinary)	OEC	OPE II(1-3)	3	0	0	3
6	V18CSL10	PCC	Advanced Java and Web Technologies Lab	0	0	2	1
7	V18CSP01	Project	Project Work (Part-A)	0	0	10	5
Total				15	0	13	21



LESSON PLANS

Advanced Java & Web Technologies

Academic Year: 2021-22

Year/ Semester: VII

Name of the Course: Advanced Java & Web Technologies

Programme: B.Tech

Section: A,B,C& D

Course Code: VI8CST27

LESSON PLAN

Course Outcomes (Along with Knowledge Level):

After completion of this course, Student will be able to:

S.No.	CO No.	Course Outcome	BTL
1.	C01	Illustrate the basic concepts of HTML and CSS	K2
2.	C02	Develop dynamic WebPages and validate with java Script.	K3
3.	C03	Illustrate Extensible markup language	K3
4.	C04	Illustrate the basic concepts of Angular JS and NODE JS.	K2
5.	C05	Build database driven web applications using JSP	K3
6.	C06	Develop web applications using PHP and MySQL	K3

Text Books:

1. Programming the World Wide Web, 7th Edition, Robert W Sebesta, Pearson, 2013.
2. Node.js, MongoDB and Angular Web Development, 2nd Edition, Brad Dayley, Brendan Dayley, Caleb Dayley, Pearson Education, 2018
3. JSP: The Complete reference, Phil Hanna, The McGraw-Hill Companies, 2001

Reference Books:

1. Web Technologies: HTML, JavaScript, PHP, Java, JSP, XML and AJAX, Black book, Dream Tech.
2. An Introduction to Web Design, Programming, Paul S Wang, Sanda S Katila, Cengage Learning
3. Web Technologies, 1st Edition 7th impression, Uttam K Roy, Oxford, 2012.

Targeted Proficiency and attainment Levels (for each Course Outcome):

Cos		C01	C02	C03	C04	C05	C06
Targeted Proficiency Level		60	60	60	60	60	60
Targeted level of Attainment	Level 3	60	60	60	60	60	60
	Level 2	55	55	55	55	55	55
	Level 1	50	50	50	50	50	50

Lecture Plan:

SNo	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO1	Dissemination of vision & mission of the department and PEO's, PO's, PSO's of the programme		1	Lecture	BB
		Describe Basic Syntax, Standard HTML Document Structure	K1	1	Lecture	BB
		Describe Basic HTML Tags	K1	1	Lecture	BB+ICT
		Explain Lists, Tables, Images	K2	2	Lecture	BB+ICT
		Explain Forms, Frames	K2	2	Lecture	BB+ICT
		Discuss Cascading style sheets, levels of style sheets and its formats	K2	2	Lecture	BB+ICT
		List Properties of Cascading style sheets	K1	3	Lecture	BB+ICT
2	CO2	Explain Objects, Primitives Operations and Expressions of JavaScript	K1	3	Lecture	BB+ICT
		Explain Output and Keyboard Input, Control Statements	K2	2	Lecture	BB
		Explain Object Creation and Modification, Arrays, Functions	K2	2	Lecture	BB
		Explain Constructors, Pattern Matching using Regular Expressions	K2	2	Lecture	BB
		Explain Events and Event handlings	K2	2	Lecture	BB
		Apply Positioning Moving with DHTML	K3	1	Lecture	BB+ICT
		Apply Changing Elements with DHTML	K3	1	Lecture	BB+ICT
3	CO3	Describe XML Syntax, XML Document Structure	K1	2	Lecture	BB+ICT
		Explain XML, Document type Definition, XML schemas	K2	3	Lecture	BB
		Demonstrate Document object model, XSLT,	K3	3	Lecture with Discussion	BB+ICT
		Illustrate DOM and SAX parsers	K3	2	Lecture with Discussion	BB+ICT
4		Show installing Node JS	K1	1	Lecture	BB+ICT

	C04	Explain Working with Node Packages, and Creating a Node	K2	1	Lecture	BB
		List JS Application, Understanding Angular, Modules, Directives,	K1	2	Lecture with Discussion	BB+ICT
		Explain Data Binding, Dependency Injection, Services	K2	3	Lecture	BB+ICT
		Explain Separation of Responsibilities, Creating a Basic Angular Application.	K2	2	Lecture with Discussion	BB+ICT
5	C05	Explain Servlets, Life cycle of Servlet,	K1	1	Lecture	BB+ICT
		List the Limitations of servlets, JSP Overview,	K2	2	Lecture	BB
		Demonstrate Components of a JSP Page: Directives, comments, Expressions, Scriptlets ,	K3	3	Lecture	BB
		Demonstrate Components of a JSP Page:Declarations, implicit objects, Database Access, session tracking	K3	3	Lecture	BB+ICT
6	C06	Overview of PHP, General syntactic characteristics,	K1	1	Lecture	BB+ICT
		Explain Primitives operations, Expressions,	K2	1	Lecture	BB+ICT
		Apply Control statements, Arrays, Functions,	K3	2	Lecture	BB+ICT
		Apply Pattern Matching, Form Handling,	K3	2	Lecture	BB+ICT
		Examine Cookies, Session Tracking	K3	2	Lecture	BB+ICT
		Develop PHP with MySQL connectivity.	K3	2	Lecture	BB+ICT

Human Computer Interaction

Academic Year: 2021-22
Year/ Semester: VII
Name of the Course: Human Computer Interaction

Programme: B.Tech
Section: A,B,C& D
Course Code:V18CST31

LESSON PLAN

Course Outcomes (Along with Knowledge Level):

After completion of this course, Student will be able to:

S.No.	CO No.	Course Outcome	BTL
1.	V18CST31.1	Describe the principles and characteristics of GUI.	K2
2.	V18CST31.2	Recognize how a computer system may be modified to include human diversity.	K2
3.	V18CST31.3	Select an effective style for a specific application.	K2
4.	V18CST31.4	Discuss Screen Designing mock-ups and carry out user and expert evaluation of interfaces.	K2
5.	V18CST31.5	Explain System Menus & Navigation Schemes	K2
6.	V18CST31.6	Discuss Device and Screen based controls.	K2

Text Books:

1. "The Essential Guide to User Interface Design", Wilbert O. Galitz, 2nd edition, 2002, Wiley India Edition.
2. Prece, Rogers, "Sharps Interaction Design", Wiley India.
3. "Designing the user interfaces". Ben Shneidermann 3rd Edition, Pearson Education Asia.

Reference Books:

1. "User Interface Design", Soren Lauesen, Pearson Education
2. "Essentials of Interaction Design", Alan Cooper, Robert Riemann, David Cronin, Wiley
3. "Human Computer Interaction", Alan Dix, Janet Fincay, GreGoryd, Abowd, Russell, Bealg, Pearson Education.

Targeted Proficiency and attainment Levels (for each Course Outcome):

Cos		CO1	CO2	CO3	CO4	CO5	CO6
Targeted Proficiency Level		70	70	70	70	70	70
Targeted level of Attainment	Level 3	70	70	70	70	70	70
	Level 2	60	60	60	60	60	60
	Level 1	50	50	50	50	50	50

Lecture Plan: UNIT-1

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
1	CO1	Dissemination of Vision, Mission, PEOs,POs,PSOs		1	Lecture	ICT
2		The User Interface: Introduction Explain Importance of the User Interface, Importance and benefits of Good Design History of Human Computer Interface	K2	2	Lecture	BB
3		List Characteristics of Graphical and Web User Interface	K1	1	Lecture with Discussion	BB
4		Describe Graphical User Interface, popularity of graphics	K2	2	Lecture	BB
5		Explain concepts of Direct Manipulation, Graphical System advantage and disadvantage	K2	2	Lecture with Discussion	BB
6		List Characteristics of GUI	K1	1	Lecture	BB
7		Illustrate Web User Interface, popularity of web, Characteristics of Web Interface	K2	2	Lecture	BB
8		Discuss Merging of Graphical Business systems& the Web	K2	2	Lecture	BB
9		List Principles of User Interface Design.	K1	1	Lecture	BB
10		Total		14		

UNIT-2

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
1	CO 2	The User Interface Design Process: Explain Obstacles and Pitfall in the development Process, Usability	K2	2	Lecture	ICT
2		Describe The Design Team, Human Interaction with Computers	K2	2	Lecture	ICT
3		List Important Human Characteristics in Design	K1	1	Lecture	ICT
4		Illustrate Human Consideration in Design, Human Interaction	K2	2	Lecture	BB

		Speeds				
5		Distinguish Performance versus Preference	K2	1	Lecture with Discussion	BB
6		Explain Methods for Gaining and Understanding of Users.	K2	2	Lecture	BB
7		Total		10		

UNIT-3

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
1	CO 3	Understanding Business Functions: Define Business Definitions & Requirement analysis	K1	2	Lecture with discussion	ICT
2		Explain Determining Business Functions	K2	2	Lecture	BB
3		Describe Design standards or Style Guides	K2	2	Lecture	BB
4		Explain System Training and Documentation	K1	1	Lecture	BB
9		Total		07		

UNIT-4

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
1	CO 4	Principles of Good Screen Design: Explain Human considerations in screen Design	K2	2	Lecture with Discussion	ICT
2		List interface design goals	K1	1	Lecture	BB
3		Discuss test for a good design	K2	2	Lecture	ICT
4		Explain screen meaning and purpose	K2	2	Lecture with Discussion	BB
5		Describe Technological considerations in Interface Design.	K1	2	Lecture	BB
6		Total		9		

UNIT-5

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
1	CO5	System Menus and Navigation Schemes: Illustrate Structure, Functions	K2	2	Lecture	ICT
2		Describe Context, Formatting	K1	1	Lecture	BB
3		Explain Phrasing and Selecting, Navigating of Menus	K2	2	Lecture with Discussion	ICT
4		List Kinds of Graphical Menus Windows Interface	K1	1	Lecture	BB
5		Discuss Windows characteristic, Components of Window	K2	2	Lecture	BB
6		Explain Windows Presentation Styles	K2	1	Lecture with Discussion	BB
7		Discuss Types of Windows	K2	1	Lecture with Discussion	ICT
8		Explain Window Management, Web systems	K2	2	Lecture	BB
9		Total		12		

UNIT-6

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
1	CO 6	Device and Screen-Based Control: Explain Device based controls, Operable Controls	K2	2	Lecture with Discussion	ICT
2		Discuss Text entry/read-Only Controls	K2	1	Lecture	BB
3		Explain Section Controls, Combining Entry/Selection Controls	K2	2	Lecture	ICT

4		Describe Other Operable Controls and Presentation Controls	K1	2	Lecture	BB
5		Illustrate Selecting proper controls	K2	1	Lecture	BB
6		Total		8		

Total No. of Classes: 60

Distributed Systems

Academic Year: 2021-22

Year/ Semester: VII

Name of the Course: Distributed Systems

Programme: B.Tech

Section: A,B,C& D

Course Code:VI8CST32

LESSON PLAN

Course Outcomes (Along with Knowledge Level):

After completion of this course, Student will be able to:

S.No.	CO No.	Course Outcome	BTL
1.	CO1	Describe distributed system and desired properties of such systems.	[K2]
2.	CO2	Discuss the theoretical concepts, namely, virtual time and agreement.	[K2]
3.	CO3	Discuss the basic theoretical concepts of distributed systems and Characteristics of IPC protocols.	[K2]
4.	CO4	Explain the mechanisms such as Remote procedure call (RPC/RMI) and OSS.	[K2]
5.	CO5	Explain the mechanisms such as file systems and P2P algorithms.	[K2]
6.	CO6	Discuss the Transactions and Replications in distributed systems.	[K2]

Text Books:

1. "Distributed Systems- Concepts and Design", George Coulouris, Jean Dollimore, Tim Kindberg, Fourth Edition, Pearson Publication
2. "Distributed Computing, Principles, Algorithms and Systems", Ajay D Kshemkalyani, MukeshSinghal, Cambridge.

Reference Books:

1. "Distributed Systems, Principles and Paradigms", Andrew S. Tanenbaum, Maarten Van Steen, 2d Edition, PHL.
2. "Distributed Systems, An Algorithm Approach," SukumarGhosh, Chapman &HalyCRC, Taylor &Fransis Group, 2007.

Targeted Proficiency and attainment Levels (for each Course Outcome):

Cos		CO1	CO2	CO3	CO4	CO5	CO6
Targeted Proficiency Level		60	60	60	60	60	60
Targeted level of Attainment	Level 3	75	70	70	65	70	65
	Level 2	70	65	65	60	65	60
	Level 1	65	60	60	55	60	55

Lecture Plan:

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 1	Dissemination of Vision, Mission, PEOs,POs,PSOs		1	Lecture	BB+ICT
2		Characterization of Distributed Systems: Introduction about Distributed System	K1	1	Lecture	BB+ICT
3		Explain about examples of Distributed System, Resource Sharing and the Web and Challenges	K2	2	Lecture with Discussion	ICT
4		System Models: Explain about Architectural Models- Software Layers	K2	2	Lecture	BB+ICT
5		Describe about System Architecture and Variations	K2	1	Lecture with Discussion	ICT
6		Describe about Interface and Objects and Design Requirements for Distributed Architecture	K2	1	Lecture	BB+ICT
7		Discuss about Fundamental Models, Interaction Model	K2	2	Lecture	BB+ICT
8		Discuss about Failure Model and Security Model	K2	1	Lecture	BB+ICT
		TOTAL		11		

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 2	Time and Global States: Introduction about clocks Events	K1	1	Lecture	BB+ICT
2		Explain about the Process States and Synchronizing Physical Clocks	K2	1	Lecture	BB+ICT
3		Discuss about Logical Time and Logical Clocks	K2	2	Lecture	BB+ICT
4		Discuss about Global States	K2	1	Discussion	ICT

5		Explain about Distributed Debugging	K2	1	Discussion	BB+ICT
6		Coordination& Agreement: Introduction about Coordination and Agreement:	K2	1	Lecture	BB+ICT
7		Explain about Distributed Mutual Exclusion	K2	1	Lecture	BB+ICT
8		Discuss about Elections, Multicast Communication	K2	1	Lecture	BB+ICT
9		Discuss about Consensus and Related Problems.	K2	1	Lecture	BB+ICT
		TOTAL		10		

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 3	Interprocess Communication: Introduction The API for the Internet Protocols.	K1	1	Lecture	BB+ICT
2		Explain about the Characteristics of Interprocess communication	K2	1	Lecture	BB+ICT
3		Discuss about Sockets, UDP Datagram Communication	K2	2	Lecture	BB+ICT
4		Discuss about TCP Stream Communication	K2	1	Discussion	ICT
5		Explain about External Data Representation and Marshalling	K2	1	Discussion	BB+ICT
6		Explain about Client Server Communication	K2	1	Lecture	BB+ICT
7		Explain about Group Communication	K2	1	Lecture	BB+ICT
8		Review Case Study: MPI	K2	2	Lecture	BB+ICT
		TOTAL		10		

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 4	Remote Invocation: Introduction	K1	1	Lecture	BB+ICT
2		Explain about Request-reply protocols	K2	1	Lecture	BB+ICT

3		Discuss about Remote Procedure Call	K2	1	Lecture	BB+ICT
4		Discuss about Events and Notifications	K2	1	Discussion With Lecture	ICT
5		Review Case Study: JAVA RMI	K2	1	Lecture	BB+ICT
6		Operating System Support: Introduction	K2	1	Lecture	BB+ICT
7		Discuss about the Operating System Layer and Protection	K2	1	Lecture	BB+ICT
8		Distinguish between Processes and Threads	K2	1	Lecture	BB+ICT
9		Explain Address Space	K2	1	Lecture	BB+ICT
10		Explain about Creation of a New Process	K2	1	Lecture	BB+ICT
11		Explain about Threads	K2	1	Lecture	BB+ICT
		TOTAL		11		

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	C05	Distributed File Systems: Introduction	K1	1	Lecture	BB+ICT
2		Explain about File Service Architecture	K2	1	Lecture	BB+ICT
3		Discuss about Peer-to-Peer Systems	K2	1	Lecture With Discussion	BB+ICT
4		Describe about Napster and its Legacy	K2	1	Lecture	BB+ICT
5		Explain about Peer-to-Peer Middleware	K2	1	Discussion	BB+ICT
6		Explain about Routing Overlays	K2	1	Lecture	BB+ICT
7		Review Case Study1: Sun Network File system	K2	1	Lecture	BB+ICT
8		Review Case Study1: Sun Network File system	K2	1	Lecture	BB+ICT
9		Review Case Study2: The Andrew File System	K2	1	Lecture	BB+ICT
				09		

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 6	Transactions& Replications: Introduction	K1	1	Lecture	BB+ICT
2		Describe about System Model and Group Communication	K2	2	Lecture	BB+ICT
3		Explain about Concurrency Control in Distributed Transactions,	K2	1	Lecture	BB+ICT
4		Discuss about Distributed Dead Locks	K2	2	Lecture	BB+ICT
5		Explain about Transaction Recovery and introduction of Replication	K2	2	Lecture	BB+ICT
6		Distinguish between Passive (Primary) Replication and Active Replication	K2	1	Lecture	BB+ICT
		TOTAL		09		

Management Science

Academic Year: 2020-21

Year/ Semester: VII

Name of the Course: Management Science

Programme: B.Tech

Section: A,B,C& D

Course Code: V18MBT52

LESSON PLAN

COURSE OUTCOMES (Along with Knowledge Level):

After completion of this course, the students will be able to:

S.No.	CO No.	Course Outcome	BTL
1.	CO 1	Understand various approaches to Management	K2
2.	CO 2	Explain the principles and practices of operations management	K2
3.	CO 3	Understand the Functions of Human Resource Management and Marketing Management	K2
4.	CO 4	Sketch the networks for better project management	K3
5.	CO 5	Understand the Concept of Strategic Management	K2
6.	CO 6	Describe the knowledge of contemporary management practices	K1

Text Books: 1. Dr. P. Vijayakumar&Dr. N. Apparao, management science, cengage.

2. Koontz &weihrich: Essential of Management, TMH 2011

Reference Books:

1. Dr. A. R. Aryasri, Management Science' TMH2011.
2. Koontz &Weihrich: 'Essentials of management' TMH2011
3. Seth &Rastogi: Global Management Systems, Cengage learning , Delhi,2011
4. Robbins: Organizational Behaviour, Pearson publications,2011
5. KanishkaBedi: Production & Operations Management, Oxford Publications,2011
6. Philip Kotler& Armstrong: Principles of Marketing, Pearsonpublications
7. BiswajitPatnaik: Human Resource Management, PHI,2011
8. Hitt and Vijaya Kumar: Starategic Management, Cengagelearning
9. PremChadha: Performance Management, Trinity Press(An imprint of Laxmi Publications Pvt. Ltd.) Delhi2015.
10. Anil Bhat&AryaKumar : Principles of Management, Oxford University Press, New Delhi,2015.

Targeted Proficiency and attainment Levels (for each Course Outcome):

Cos	CO1	CO2	CO3	CO4	CO5	CO6
Targeted Proficiency Level	60	60	60	60	60	60
Targeted level of Attainment	60	60	60	60	60	60

Lecture Plan:

UNIT I Introduction to Management:

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Reqd.	Pedagogy	Teaching aids
1.	CO1	Definitions of management	K1	1	Lecture	BB
2.		Concept of management	K1	1	Lecture	BB
3.		Nature and importance of Management	K1	2	Lecture	BB
4.		Describe the Functions of management.	K1	2	Lecture	BB
5.		Evaluation of management thought	K2	2	Lecture+ discussion	BB
6.		Explain Theories of motivation.	K2	2	Lecture	BB
7.		Decision-making process	K2	1	Lecture + discussion	BB
		Total		11		

UNIT II Operations Management:

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Reqd.	Pedagogy	Teaching aids
1.	CO2	Describe the Principles and types of management	K1	1	Lecture	BB
2.		Work Study and Statistical Quality Control.	K2	2	Lecture + discussion	BB
3.		Control charts (P- chart, R chart, and C chart)	K3	2	Lecture + discussion	BB
4.		Explain the concept of Material Management	K2	1	Lecture + discussion	BB
5.		Need for inventory control	K2	2	Lecture + discussion	BB
6.		EOQ, ABC analysis simple problems and Types of ABC analysis.	K3	2	Lecture + discussion	BB
		Total		10		

UNIT III Functional Management:

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Reqd.	Pedagogy	Teaching aids
1.	CO3	Concept of HRM, HRD and PMIR	K1	1	Lecture	BB
2.		Functions of HR manager	K2	2	Lecture + discussion	BB
3.		Wage payment plans	K2	2	Lecture + discussion	BB
4.		Job Evaluation and Merit Rating	K2	1	Lecture +	BB+PP

		methods.			discussion and In-class Assignment	T
5.		Marketing Management based on product life cycle.	K2	1	Lecture + discussion	BB
6.		Functions of Marketing	K2	1	Lecture + discussion	BB
7.		Four P's , Product, Price, Place and Promotion	K2	2	Lecture + discussion and In-class Assignment	BB+PP T
8		New product development	K2	1		BB+PP T
9		servicesmarketing	K2	1		BB+PP
		Total		12	10	

UNIT IV Project Management:

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Req'd.	Pedagogy	Teaching aids
1.	CO4	Development of Network	K2	3	Lecture+ discussion	BB
2.		Difference between PERT and CPM	K2	2	Lecture + discussion and In-class Assignment	BB+P PT
3.		Identifying Critical Path- Project Crashing (Simple Problems).	K3	4	Lecture + discussion	BB
		Total		09		

UNIT V Strategic Management:

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Req'd.	Pedagogy	Teaching aids
1.	C05	Describe Vision, Mission, Goals and Strategy.	K2	2	Lecture + discussion	BB
2.		Describe Strategic management process	K2	2	Lecture + discussion	BB
3.		Elements of corporate planning process.	K2	2	Lecture + discussion and In-class Assignment	BB+ PPT
		Environmental Scanning	K2	1	Lecture	BB
		SWOT analysis.	K2	1	Lecture + discussion and In-class Assignment	BB+ PPT
		Total		08		

UNIT VI Contemporary Management Practices:

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Req d.	Pedagogy	Teaching aids
1.	CO6	Describe the concept of Total Quality Management	K2	1	Lecture + discussion	BB
2.		Describe the concept of Six sigma	K2	2	Lecture + discussion	BB
3.		Describe the concept of Supply chain Management.	K2	1	Lecture + discussion and In-class Assignment	BB+ PPT
4.		Describe the concept of Business process out sourcing	K1	2	Lecture	BB
5.		Describe the concept of Lean start ups	K2	2	Lecture + discussion	BB
6.		Describe the concept of entrepreneurship	K1	2	Lecture	BB
		Total		10		

Total No. of classes: 60

Environmental Pollution and Control

Academic Year: 2021-22

Year/ Semester: V

Programme: B.Tech

Section: A,B,C& D

Name of the Course: Environmental Pollution and Control Course Code: V18CEO E03

LESSON PLAN

COURSE OUTCOMES (Along with Knowledge Level):

After completion of this course, the students will be able to:

CO1	Describe about air pollution and its control methods to students	K2
CO2	Develop the student to understand about industrial wastewater and ways to control it	K3
CO3	Describe student to understand about solid waste and methods to control it	K2
CO4	Express to student about importance of Environmental sanitation	K2
CO5	Prepare student to understand about Hazardous waste and ways to control it	K3
CO6	Illustrate the importance of Sustainable development to student	K3

Text Books:

1. Environmental Engineering, by Ruth F. Weiner and Robin Matthews – 4th Edition Elsevier, 2003.
2. Environmental Science and Engineering by J.G. Henry and G.W. Heinke – Pearson Education.
3. Environmental Engineering by Mackenzie L Davis & David A Cornwell. McGraw Hill Publishing.

References:

1. Air Pollution and Control by M.N. Rao & H.N. Rao
2. Solid Waste Management by K. Sasi Kumar, S.A. Gopi Krishna. PHI New Delhi.
3. Environmental Engineering by Gerard Kiley, Tata McGraw Hill.
4. Environmental Sanitation by KVSG Murali Krishna, Reem Publications, New Delhi.
5. Industrial Water Pollution Control by Nemerow Jr., McGraw Hill Publishing.
6. Unit Operations and Processes in Environmental Engineering by Reynolds. Richard – Cengage Learning.
7. Environmental Engineering by D. Srinivasan, PHI Learning Private Limited, New Delhi, 2011.
8. Environmental Engineering – Howard S. Peavy, Donald R. Rowe, George Tchobanoglous – Mc-Graw-Hill Book Company, New Delhi, 1985.

Course out come	Targeted Proficiency Level	Targeted level of Attainment
CO1	60%	60%
CO2	60%	60%
CO3	60%	60%
CO4	60%	60%
CO5	60%	60%
CO6	60%	60%

<u>Sno</u>	<u>Course Outcome</u>	<u>Knowledge Level</u>	<u>Contents</u>	<u>No. of Hours Required</u>	<u>Pedagogy</u>	<u>Teaching aids</u>
1	CO1	K2	Explain AirpollutionControlMethods	2	Lecture with discussion	Black Board
		K1	Explain Particulate controldevices	3	Lecture with discussion	Black Board
		K2	Classify MethodsofControllingGaseous Emissions	2	Lecture with discussion	Black Board
		K1	List Airqualitystandards	1	Lecture with discussion	Black Board
		K2	Explain NoisePollution	1	Lecture with discussion	Black Board
		K1	List NoiseStandards	1	Lecture with discussion	Black Board
		K2	Discuss Measurement andcontrolmethods	1	Lecture with discussion	Black Board
		K2	Discuss Reducingresidentialandindustr alnoise	2	Lecture with discussion	Black Board
		K2	Discuss Reducingresidentialandindustr alnoise	2	Lecture with discussion	Black Board
		K2	Discuss ISO14000	1	Lecture with discussion	Black Board
2	CO2	K1	Describe Industrial wastewater Management interdiction	1	Lecture with discussion	Black Board

		K2	Explain Volume and Strength reduction	2	Lecture with discussion	Black Board
		K3	Solve problems on Volume and Strength reduction	2	Lecture with discussion	Black Board
		K2	Discuss Strategies for pollution control	1	Lecture with discussion	Power point presentation
		K1	Define Equalization, Neutralization and Proportioning	2	Lecture with discussion	Black Board
		K2	Discuss Common Effluent Treatment Plants	1	Lecture with discussion	Black Board
		K2	Explain Recirculation of industrial wastes	2	Lecture with discussion	Black Board
		K1	List Effluent standards	1	Lecture with discussion	Black Board
3	CO3	K2	Discuss Solid Waste Management interdiction	1	Lecture with discussion	Black Board
		K2	Discuss solid waste characteristics	2	Lecture with discussion	Black Board
		K2	Explain basics of on-site handling and collection	1	Lecture with discussion	Black Board
		K2	Discuss separation and processing	1	Lecture with discussion	Black Board
		K2	Explain Incineration-Composting	2	Lecture with discussion	Black Board
		K2	Discuss Solid waste disposal methods	1	Lecture with discussion	Power point presentation
		K2	Explain Fundamentals of Land filling	1	Lecture with discussion	Black Board
4	CO4	K1	Explain Environmental Sanitation interdiction	1	By lecture	Black Board
		K2	Explain Environmental Sanitation Methods for Hotels	3	By lecture	Black Board
		K2	Explain Environmental Sanitation Methods for	2	By lecture	Black Board

			Hospitals			
		K2	Explain Environmental Sanitation Methods for Swimming pools	2	By lecture	Black Board
		K2	Explain Environmental Sanitation Methods for public bathing places	2	By lecture	Power point presentation
		K2	Explain Environmental Sanitation Methods for social gatherings (melas and fares)	2	By lecture	Black Board
		K2	Explain Environmental Sanitation Methods for Schools and Institutions	2	By lecture	Black Board
		K2	Explain Environmental Sanitation Methods for Rural Sanitation	2	By lecture	Black Board
		K2	Explain Environmental Sanitation Methods for low cost waste disposal methods	2	By lecture	Black Board
5	CO5	K2	Discuss Hazardous Waste interdiction	1	By lecture	Black Board
		K2	Explain Characterization	2	By lecture	Power point presentation
		K2	Discuss Nuclear waste, Biomedical wastes and Electronic wastes	3	By lecture	Black Board
		K2	Discuss Chemical wastes	1	By lecture	Black Board
		K2	Explain Treatment and management of hazardous waste	2	By lecture	Black Board
		K2	Discuss Disposal and Control methods.	2	By lecture	Black Board
		K1	Define Sustainable Development and interdiction	2	By lecture	Power point presentation

6	CO6	K2	Explain elements of sustainable developments	3	By lecture	Black Board
		K2	Discuss Indicators of sustainable development	2	By lecture	Black Board
		K2	Explain Sustainability Strategies	1	By lecture	Black Board
		K2	Discuss Barriers to Sustainability	1	By lecture	Black Board
		K2	Explain Industrialization and sustainable development	2	By lecture	Black Board
		K2	Explain Discuss Cleaner production in achieving sustainability, sustainable development		By lecture	Black Board

Advanced Java and Web Technologies Lab

Academic Year: 2021-22

Year/ Semester: VII

Name of the Course: Advanced Java and Web Technologies Lab

CourseCode:V18CSL10

Programme: B.Tech

Section: A,B,C& D

LESSON PLAN

COURSE OUTCOMES (Along with Knowledge Level):

After completion of this course, the students will be able to:

CO1	Develop static web pages using HTML, CSS.	[K3]
CO2	Demonstrate the concepts of JavaScript and DHTML.	[K2]
CO3	Demonstrate the basic concepts of Angular JS and NODE JS.	[K2]
CO4	Demonstrate the concepts of Extensible markup language & AJAX	[K2]
CO5	Develop Web Applications using JSP.	[K3]
CO6	Develop dynamic Web Applications using PHP & MySQL	[K3]

Text Books:

1. Programming the World Wide Web, 7th Edition, Robert W Sebesta, Pearson, 2013.
2. Node.js, MongoDB and Angular Web Development, 2nd Edition, Brad Dayley
Brendan Dayley Caleb Dayley, Pearson Education, 2018
3. JSP: The Complete reference, Phil Hanna, The McGraw-Hill Companies, 2001

Targeted Proficiency and Attainment Levels (for each course Outcome):

Cos		CO1	CO2	CO3	CO4	CO5	CO6
Targeted Proficiency Level		75	75	75	75	75	75
Targeted level of Attainment	Level 3	70	70	70	70	70	70
	Level 2	65	65	65	65	65	65
	Level 1	55	55	55	55	55	55

Lecture Plan:

S.NO	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours required	Pedagogy	Teaching aids
1	CO 1	Dissemination of Department Vision, Mission, PEOs, POs, PSOs	-	-	-	
		Design the following static web pages required for an online book store web site: (a) HOME PAGE: (b) LOGIN PAGE:	K3	3	Lecture With Discussion	ICT
		Design the following static web pages required for an online book store web site: (c) CATALOGUE PAGE: (d) REGISTRATION PAGE:	K3	3	Lecture With Discussion	ICT
3		Design a web page using CSS (Cascading Style Sheets) which includes the following: Use different font, styles: In the style definition you define how each selector should work (font, color etc.).	K3	3	Lecture With Discussion	ICT

S.No	Course Outcome	Intended Learning Outcomes(ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 2	Design a login page and Make use of Events to perform validation using JavaScript.	K2	3	Lecture with Discussion	ICT
2		Demonstrate a JavaScript program to perform On Mouse over event	K2	3	Lecture with Discussion	ICT

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 3	Demonstrate the concept of Mouse events (Ex:ng-click) with the help of Angular JS.	K2	3	Lecture with Discussion	ICT
2		Design a simple Angular JS form.	K2	3	Lecture with Discussion	ICT

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 4	Write an XML file which will display the Book information which includes the following: 1) Title of the book 2) Author Name 3) ISBN number 4) Publisher name 5) Edition 6) Price	K2	2	Lecture with Discussion	ICT
2		a) Write a Document Type Definition (DTD) to validate the above XML file. b) Write a XML Schema Definition (XSD)	K2	1	Lecture with Discussion	ICT

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 5	Create a simple JSP to print the current Date and Time.	K3	2	Lecture with Discussion	ICT
2		Create JSP to insert the details of 3 or 4 users using a registration form store these values in the data base and then check the authentication of the user by entering the name and password using a login form.	K3	1	Lecture with Discussion	ICT

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 6	Assume four users user1, user2, user3 and user4 having the passwords pwd1, pwd2, pwd3 and pwd4 respectively. Write a PHP for doing the following. a) Create a Cookie and add these four user id's and passwords to this Cookie. 2. Read the user id and passwords entered in the Login form (week1) and authenticate with the values (user id and passwords) available	K3	3	Lecture with Discussion	ICT

		in the cookies. If he is a valid user (i.e., user-name and password match) you should welcome him by name (user-name) else you should display “You are not an authenticated user ”. B) Use init-parameters to do the same.				
2		Create a table which should contain at least the following fields: name, password, email id, phone number (these should hold the data from the registration form). Write a PHP program to connect to that database and extract data from the tables and display them. Experiment with various SQL queries. Insert the details of the users who register with the web site, whenever a new user clicks the submit button in the registration page.	K3	3	Lecture with Discussion	ICT
3		Insert the details of the 3 or 4 users who register with the web site by using registration form. Authenticate the user when he submits the login form using the user name and password from the database.	K3	3	Lecture with Discussion	ICT

Total No. of Classes: 36